

NeurTEx: A Neural Framework for Template Extraction from Flat Images

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ABSTRACT

With the increasing demands of digital creation, a creative user often starts by looking for inspirational resources online. Once they find such inspirations, they recreate certain design elements using tracing-like applications to internalize the design for their purpose. We aim to accelerate this process by extracting key semantics from an inspirational image, and converting it into a ready-to-use template using a multimodal information extraction setup. We propose **NeurTEx**, a holistic algorithm that takes an inspirational banner image as input and extracts multimodal design semantics: layout, text elements (actual text along with the font style), image elements (including semantics like logos and background/foreground objects) and shapes. Our technique uses a segmentation framework followed by a region based depth-first search to extract and identify different elements and their semantics. We process these regions to extract finer details such as font styles, texts and logos. We believe that such fine-grained semantics would accelerate the design process for a creator helping them rapidly adapt designs from inspirations for their requirements. With the help of metric-based and human survey based evaluations, we not only demonstrate the effectiveness of the proposed approach to extract the style & design components from an inspirational image but also illustrate how these extractions can accelerate the creation process thus aiding novices and professionals alike.

CCS CONCEPTS

• Computing methodologies → Image processing; • Information systems → Multimedia content creation.

KEYWORDS

layout extraction, single page graphic design, automated systems

*Work done when authors were at Adobe Research

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1 INTRODUCTION

Designing aesthetically pleasing banners that appeal to the target recipients is a non-trivial task and requires considerable human effort and time. Once created, the design of these banners influences customers' interest and plays an important role in optimizing the click-through rates of the digital banners [40]. A banner is a graphic design that can be thought of as an ensemble of an intent, a theme, a collection of images, affiliated text phrases (in a specific font style - typography, size, and color), and additional elements including logos, shapes, and textures [21, 40]. Typically, designers start by scouting for inspirational banners and begin their creation by mimicking certain aspects of the design from the inspiration [22]. Replicating or mimicking the inspirational design is non-trivial and requires extensive knowledge of creative tools such as ¹ to recreate or extract relevant style information from the inspirational banners.

An automated pipeline to extract the key design semantics from a flat banner image into a form that can be readily adapted by a designer for a new creation is valuable and saves a lot of time and effort. An inspirational "**flat banner image**" is a digital image with no additional metadata present, found on a website or on social media posts. We propose **NeurTEx**, a framework to automatically extract key design semantics (text, image and shapes) from a multimodal banner image along with their layout to arrive at the underlying template. We further enrich this template by extracting fine-grained semantic information from the image and text elements via logo detection, font detection and text detection. We believe the template thus extracted aids in accelerating the creation process contributing in quick conversion of inspirations to a usable designs - as also validated in our human evaluation. Key contributions of this work are:

- We propose **NeurTEx**, a novel pipeline to extract the layout and other semantic information from flat banner images.

¹<https://www.adobe.com/in/products/photoshop.html>



Figure 1: From left → right: Example depicting outputs at various stages of the proposed NeurTex pipeline - Banner Image; Semantic Mask; Extracted Layout; Text Segments; Identified Fonts; Identified Text; Identified Logo

- We extend existing works in logo detection, font detection and text extraction to build a comprehensive pipeline for fine-grained design semantics extraction.
- Through extensive ablations and human survey, we show the viability of the proposed pipeline.

2 RELATED WORK

Traditionally information extraction from images have focused on document images. Toledo et al. [38] tackle the problem of extracting text content from digital document images. Paliwal et al. [25] propose a framework for extracting tabular structure from scanned document images. Recent advancements in deep neural networks have led to the exploration of novel information extraction tasks in the image space. Beltramelli et al. [3] proposed pix2code, aimed at extracting layout information from screenshots using an RNN-based architecture. Maheshwari et al. [21] extracted and optimized layout from a banner image using an energy-based approach. In the color domain, works such as Grogan et al. & Kumar et al. [10, 14] extract color palettes from images using methods such as K-means clustering and grouping of colors while preserving diversity. However, these works have focused on extracting a single aspect from the input image; whereas in NeurTex, we propose a comprehensive pipeline consisting of a cohort of graphic design tools, that extracts multiple aspects from a digital banner image at various granularities - a key distinction from the existing works.

Works on **image segmentation** and **object detection** are also closely related to information extraction from images. The advent of Fully convolution networks (FCN) [19] have triggered development of DeconvNet [23], SegNet [1] and RefineNet [15] - approaches that adopted the encoder-decoder structure on top of the FCNs, to combine low and high level information for optimized segmentation. In object detection, YOLO and its variants [27–29] have been successfully deployed for multiple object detection tasks. Earlier, He et al. [11] proposed RCNN based object detection architecture while Liu et al. [18] proposed a single shot object detection model that discretizes the output space of bounding boxes into a set of default boxes over different aspect ratios and scales for a feature map location. In NeurTex, we compare & evaluate both these classes of frameworks for extracting the design semantics from flat images, and leverage segmentation based framework due to their superior performances for design extraction.

While several approaches exist in the field of image understanding and some scattered approaches exist in the creative design field, to the best of our knowledge none of these provide a comprehensive approach to extract deeper semantics of various design elements present in banner images. This work leverages and enhances the existing approaches to deliver a seamless experience by extracting



Figure 2: Sample Banner Images from Graphic Template Dataset

a wide array of style and design aspects from the inspirational banner image and making them reusable to accelerate the workflow of designers to enable creativity for all.

3 GRAPHIC TEMPLATE DATASET

All the experiments are done on Graphic Template Dataset comprising of 140k creative designs consisting of posters, flyers, collages, social media posts, ads and more. It has been tagged for different regions of interest, font styles & color information. The dataset is proprietary and is curated internally for experimentation on large scale.

4 NEURTEX: OVERVIEW

A banner is designed to capture an underlying theme. It is an expression of the designer’s creativity to incorporate the message into different visual and textual elements and therefore the designing process is inherently complex. Figure 2 shows sample banners present in our proprietary dataset (Graphic Template Dataset) of 140k banners. The elements constituting the banner, dominantly images and textual phrases, can be arranged on the banner canvas in multiple ways. This arrangement is dubbed as *Layout* of the banner. Extracting the layout from a flat image becomes a non-trivial task due to the presence of multiple modalities and style aspects. The presence of additional elements such as *shapes* that are placed on the canvas so that the text can be overlaid on top of them (Figure 2c), instead of the background image itself (Fig. 2b), makes the problem of identifying key regions of a banner even more intricate.

Beyond the regions of interest (ROIs), other semantics associated with the elements of the banner such as the font typography, logos are also important for a designer. Extracting such details requires very different architectures and techniques. Thus, we propose NeurTex, a multi-stage pipeline to identify and extract design semantics from a flat banner image into a readily usable format. Fig. 3 shows the sequence of steps in NeurTex for coarse and fine grained design semantic extraction from an input image. We now describe different modules of NeurTex in detail.

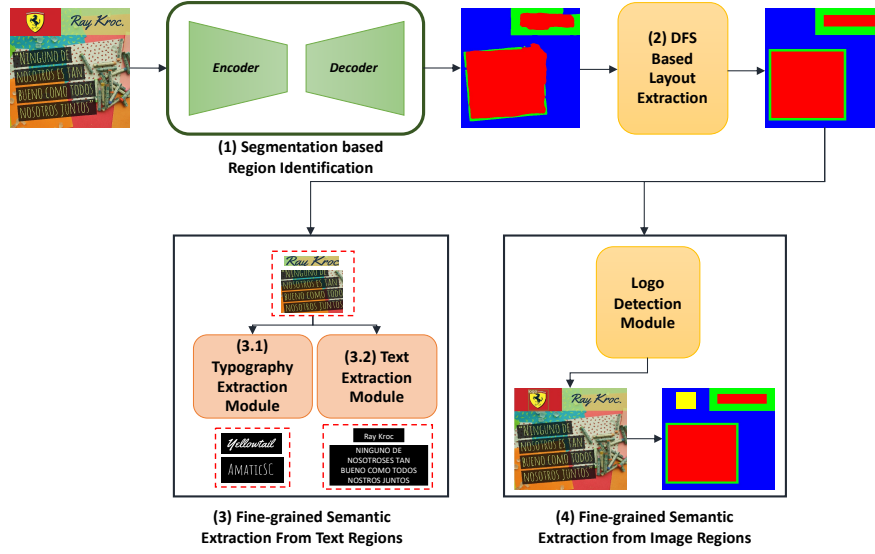


Figure 3: NeurTex pipeline to extract Semantic Information from Banners

4.1 Layout Extraction & Region Identification

Starting with an input image, NeurTex first identifies different *elements* in the input image along with a rectangular bounding box indicating their position and size in the layout. This requires the detection of different regions, categorization of the identified elements into different element classes, and identifying the bounding boxes in the image to arrive at the template. Layout extraction in NeurTex entails 2 key questions: (1) where are the elements present in a given image?, (2) what are the semantics of the elements present at the identified regions?

In our dataset, there are 3 key regions of a banner - *image, text and shape*. Each banner can have multiple instances of each of the three classes. For e.g, Fig. 2a contains 2 images, 2 text regions and each text segment rendered on separate shapes; Fig. 2b has 1 image, 1 text and no shape; Fig. 2c has 1 image, 2 texts and each text on separate shapes.

One possible way to extract layout semantics is to cast it as a **semantic segmentation** task and segment each pixel of the banner into one of the 3 classes - image, text and shape. An alternative is to formulate the problem as an **object detection & localization** and localize the regions of the images which qualifies into these 3 categories and obtain the rectangular coordinates for each class element. We evaluate both these formulations and arrive at an optimal choice for NeurTex. Note that, unlike object detection, the segmentation approach would need an additional step to arrive at concrete regions from the pixel-level semantics..

Semantic Segmentation based Layout Extraction: Recent segmentation frameworks are built on top of Fully Convolutional Neural Networks (FCNN) [19], with an encoder-decoder setup where the task of the encoder is to continuously down-sample (contracting) the original image and learn low-level features from it. The output of each down-sampling layer is utilized in the decoding stage. Since the output segmentation mask should be of the same

size as the input image, the decoder is responsible for learning the masks as well as for up-sampling the final encoder output. The stored output of each down-sampling convolution layer is passed as input to the corresponding up-sampling layer. We adopt the same mechanism for our framework, which consists of an **encoder** that acts as feature extractor (the backbone), a **decoder** that acts as feature fusion block to create segmentation masks and a final **segmentation head** to reduce number of channels from decoder and up sample mask to preserve input-output spatial resolution identity.

Our segmentation setup comprises of ResNet50 [12] loaded with weights trained on Imagenet [6] as the encoder. Using such pre-trained encoders provides additional knowledge to the model of the kind of image features that need to be detected and improved. Residual Networks, or ResNets[12], learn residual functions with reference to the layer inputs, instead of learning unreferenced functions. Instead of hoping each few stacked layers directly fit a desired underlying mapping, residual nets let these layers fit a residual mapping. They stack residual blocks on top of each other to form network: e.g. a ResNet-50 has fifty layers using these blocks.

For decoders, we experimented with U-NET and Feature Pyramid Network as they have shown remarkable performances on segmentation datasets like COCO [17]. U-net [30] is composed of two parts: a contracting part to compute features and a expanding part to spatially localise patterns in the image. The downsampling or contracting part has a FCN-like architecture extracting features with 3×3 convolutions. The upsampling or expanding part uses up-convolution (or deconvolution) reducing the number of feature maps while increasing their height and width. Feature Pyramid Network (FPN) [16] is an extension of U-net with significantly better inference speeds. Its architecture is composed of a bottom-up pathway, a top-down pathway and lateral connections in order to join low-resolution and high-resolution features. Note that our aim

is not to introduce a novel segmentation model but to leverage existing architectures for extracting layout semantics from flat banner images.

To arrive at the boundaries of different regions from the pixel-wise segmentation (going from second to third image in 1), we employ a Depth First Search (DFS) based method which uses connected components to process the segmentation output and extract the positional coordinates and size of the underlying elements. We start traversing the segmented output and for each pixel, the algorithm starts a DFS to find the largest rectangular box that can have the initial pixel as one of its corners. Since the bounding box search does not limit the selection of a pixel in multiple elements, this allows for overlapping elements to be identified. Bounding boxes less than a threshold (in size) were rejected to control the noise in semantic segmentation. The approach is inspired from the work done by [21].

We trained each encoder-decoder setup on the graphic template dataset (Sec 3) using a train, validation and test split of 8:1:1. We trained for 5 epochs with the training & validation scores logged after every 400 steps. The learning rate was kept as $1e-4$ throughout the training and the batch size was set as 16. The results are summarized in Table 1a.

Object Detection based Layout Extraction: YOLO [27] has been the center of object detection methods for the past few years outperforming other detection methods such as DPM and R-CNN [11]. YOLOv3 [29] is the enhanced version of the original YOLO algorithm in both accuracy and speed. It implements Darknet-53 architecture as its backbone with three various scales for detection also named as “output layers” or “heads”. One upside of using an object detection setup for layout extraction is it decouples the task of mask-identification from region-classification leading to smooth rectangular boundaries for each detected region. We train a standard YOLOv3 model to extract the layout and identify key regions from the banner image. The same train, val and test split is used as the segmentation experiments to compare which method outperforms the other.

Segmentation vs Detection: To evaluate layout semantic extraction, we use three metrics. *Intersection over Union (IoU)*, also referred as *Jaccard Index*, quantifies the percent overlap between the ground truth mask/regions against the prediction output. For segmentation models, IoU metric measures the number of pixels common between the target and prediction masks divided by the total number of pixels present across both masks. For object detection models, IoU measures the intersection area overlap between two rectangular boxes divided by the union of the areas occupied by the two boxes. *Precision* and *Recall* values are calculated on top of IoU scores using a threshold of 0.5, i.e. a detection which has an IoU score ≥ 0.5 is considered as a positive detection.

We observe from Table 1a that both the segmentation setups outperforms the object-detection based setup considerably. Interestingly, Unet and Feature Pyramid Network (FPN) saturated at approximately the same values. From the precision and recall values for layout extraction, it can be deduced that the segmentation technique is more suitable for the task at hand allowing identification of more succinct and unconventional elements. The DFS based post processing can be extended to accommodate complex

Model	IoU Score	Precision	Recall
Resnet-Unet	0.7969	0.8855	0.8855
Resnet-FPN	0.8032	0.8895	0.8893
YOLO V3	0.6024	0.5825	0.6143

(a) Results for Layout Extraction and Region Identification

Model	ResNext50 _32x4d [44]	Resnet50 [12]
Avg. Macro Precision	0.3930	0.4377
Avg. Macro Recall	0.5407	0.6127
Avg. Macro F-Score	0.4113	0.4678
Micro Prec.	0.4733	0.5178

(b) Results for Font Identification from text segments

Table 1: Results from different parts of NeurTex

boundary structures present in the banner image and is a subject of further work. Such a flexibility, however is missing in object detection and classification setup, which is limited to rectangular bounding boxes. We, therefore, use Resnet-FPN based segmentation setup for layout extraction in NeurTex for further considerations as they outperform the other two.

4.2 Extracting Deeper Semantics

Typography Recognition: Visual Font Recognition (VFR) is the task of classifying an input character image into one of the registered font classes. Prior to the deep learning based models, a number of systems, with the objective of improving the OCR engines, leveraged structure and appearance based features [9], local feature embeddings [5], gabor and gradient features [20, 26], and representative stroke templates [37] to identify the fonts. One of the first deep learning architectures for font recognition, the DeepFont model [43], successfully used the CNNs and stacked convolutional auto-encoders to improve the recognition and domain adaptation tasks respectively.

In NeurTex, we formulate the typography/font identification from textual segments as a classification task inspired by the work of Wang et al. [43]. The text regions are cropped from the training images and used to train the model (195k textual segments falling under 49 different font categories). We experimented with Resnet50 [12] and ResNext50 [44], both pre-trained on ImageNet dataset[6]. We trained for 30 epochs with a batch-size of 256 and a constant learning rate of $5e-4$ was used. Each text segment in training set is transformed by applying random cropping, rotation, horizontal flip and color jitters to make the model robust during the testing phase. No transformation is applied to validation set except center cropping and resizing. Table 1b shows the accuracy, precision, recall and f-score values for each architecture on the held out test set consisting of 22k text segments. We observe that Resnet [12] outperforms the other architecture. Hence, it becomes the natural choice to be used as the font identification module in NeurTex.

Text Identification: Identifying text from images is an age old problem. Recognizing contents of a document image, identifying the language of the text and its size is of great use. Optical Character Recognition (OCR) techniques are widely used for electronically converting images of typed, handwritten or printed text to machine-encoded text. Tesseract [32] is an open source text recognition engine released by Google and is trained using LSTMs to recognize character patterns. It is widely used to OCR digital documents and

images. We have employed it for identifying the contents of the textual regions in NeurTex. However, it cannot be directly used for banner images because the text present in banners can be *warped, stretched, resized or skewed* and therefore very divergent from the kind of data the engine has been trained on. To overcome this, we take each textual segment, cropped from the image, place it on a white background and resize the textual segment to one-eighth the size of white background. This ensures that the size of the text isn't huge. White background is used rather than the original banner image to ensure no external noise affects the performance of the engine. Running OCR on such a proxy image leads to a robust starting point resulting in better text recognition in our experiments.

Logo Detection: Brands often design banners to target their customers and one key feature of such banners is the presence of brand logos. Detecting the presence of these logos and localizing their position in the banner image add an additional layer of information which can be utilized by designers for easier customization. Availability of large scale datasets [4, 13, 36, 41] that cater to different needs made the logo detection task suitable for learning using deep neural networks. Several works [4, 24, 31, 41] have transformed the object detection techniques such as RCNN, Faster RCNN, SSD, YOLO, RFBNet, etc., to identify logos in images, thus making the logo detection task a special case of the generic object detection task. To determine the level of categorization possible and tackle the issue of scalability in recognizing the logos, researchers have developed techniques such as self-co-learning [33, 35], proxies [7], and open-set logo detection [2, 36, 39] among many others [8, 34, 42].

We train a YOLOv3 Object Detection model [29] using Logo-32[24] dataset but formulate the detection problem as an *instance identification problem*, viz. rather than training the model for detecting each brand logo separately, we use a single class "logo" to be detected within the image regions. An image can have 0 or more logos and our logo detection model is trained to predict the position of all such logo candidates. Note that although in theory, logo detection can be clubbed with the layout detection task, the Graphic Template Dataset does not have this information tagged. Hence, a separate object detection model is trained on the openly available Logo-32 dataset and is added to the overall pipeline for completion. We partitioned the Logo-32[24] dataset as 8 : 1 : 1 train-validation-test split. We use a pretrained DarkNet53 model as backend and fine-tuned it on Logo-32 dataset for 300 epochs. The model is able to recognize logos in the image with a 75% recall, 51% precision and 0.6 F1.

5 NEURTEX FOR REPURPOSING DESIGN ELEMENTS

In previous sections, we discuss the statistical evaluation of extracting the layout and style aspects from a banner using the proposed NeurTex pipeline, as reported in Tables 1a and 1b. Since there is no off-the-shelf criterion to evaluate the complete pipeline together, we provide independent evaluations for each of the sub-modules. However, the intended goal of our pipeline is to facilitate the designers in creating new banners on the fly by taking inspiration from another banner. To facilitate an end-to-end evaluation and to understand whether the extracted information about the layout

structure, location of each element, font typography, and textual content can actually help in creating visually appealing new banners, we conduct a human evaluation.

As the first step of our evaluation process, we first select a random set of banners created by designers (*human generated*) and pass them through the NeurTex pipeline to extract the layout and semantic information. Then, we randomly select new background images and use the extracted information to create a set of synthetic banners. The text position in the synthetic banners is kept the same as it was in the human designed banners.

An annotator is first shown a banner image which can come either from the set of banners created by designers (set A) or the set of banners created synthetically using NeurTex (set B). We randomly show an image from either of the sets (A) or (B) and ask the annotator to score the image on a Likert scale (1 – 5) on three dimensions: (a) how aesthetically pleasing the banner image is; (b) how much would they prefer using the banner for some creative purpose; (c) how well the banner image is able to convey the intended message. We call these ratings as *Individual Ratings* of the human and machine generated banners. We do not expose the label of the banners (synthetic or human generated) and randomize the order of their presentation to the annotator to avoid any preferential biases in scoring the images. Additionally, we also ask the annotator to give a binary choice between the two banners on the same three dimensions (a) which image is more aesthetically pleasing; (b) which one is more preferred for some creative purpose; (c) which banner is better able to convey the intended message. We call these as *Overall Ratings*. We have designed these questions to check the value added by NeurTex in the end-to-end banner creation process. High scores are desirable for all the three dimensions.

For the **individual ratings**, we obtained 100 annotations per set using the MTurk (<https://www.mturk.com/>) platform. Figure 5 shows the fraction of annotations that received a specific rating across the three dimensions through histogram plots. A large fraction of human generated banners get a rating of 4 or 5, which is expected. However, we observe that a large fraction of the banners generated using NeurTex also received a rating of 3 and 4, indicating that the proposed extraction eases the generation of new designs from the extracted templates. Note that, we do not expect the machine generated banners to get very high ratings, since the designer would still work on top of the extracted semantics to finesse the design, which did not happen in our setup. The **avg. aesthetics score, preference score & message conveyance score** for the human generated banners and synthetically generated banners were (3.79, 3.69), (3.59, 3.62) and (3.75, 3.60) respectively. We can see that the scores are comparable for both the modes of banner creation further showing the viability of NeurTex to extract designs with a high fidelity. In the pair-wise comparisons, we found that 51%, 52% and 48% preferred human-generated banners over the NeurTex generated banners on the scale of aesthetics, preference and message conveyance respectively. The fact that the preference value was ~ 50% again indicates the usefulness of extracted templates to accelerate creation.



Figure 4: Example Banners created by extracting information using NeurTeX and combining the info with new image & text segments

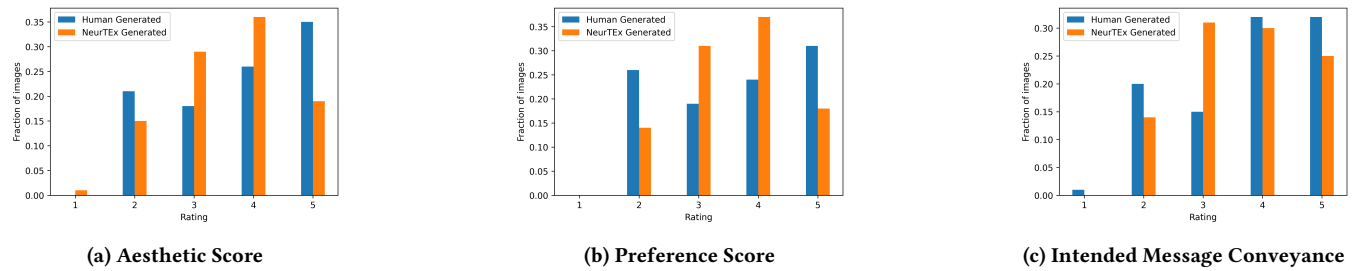


Figure 5: Comparing various scores for NeurTeX and human generated banners

6 CONCLUSION & FUTURE WORK

Understanding and analyzing multimedia content has been a popular topic of interest. We propose the NeurTeX framework as a tool for interpreting, analyzing, and extracting information from multi-modal input such as banner designs. Here, we study the problem of layout and style information extraction from an inspirational banner image to assist the creative professionals in designing new banners by accelerating their workflow. We showed that the problem comprises of significantly different sub-tasks and requires a cascaded pipeline for an end-to-end solution. Our algorithm uses a semantic segmentation based region identification followed by a Depth First Search based layout extraction for identifying and localizing key regions. The textual regions are processed to identify the font typography and content information while the image regions are processed to identify the logos in the foreground. We hope our work will pave way for more research on extracting information from multi-modal inputs and storing it structurally to aid the creation process by assisting the creative designers. The current setup is limited to identifying background image, shapes and textual regions. A dataset consisting of a broader list of foreground and background objects could be invaluable in extending the work. We will work towards extending the Graphic Template Dataset and get more things annotated for further training. We also plan to extend our work by extracting semantic information from a collection of banners and auto-remixing those components intelligently to come up with multiple ready-to-use banner variants that can serve as a starting point for a new creation. We hope such automated pipeline will help in accelerating and assisting the designers as well as hobbyist users.

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